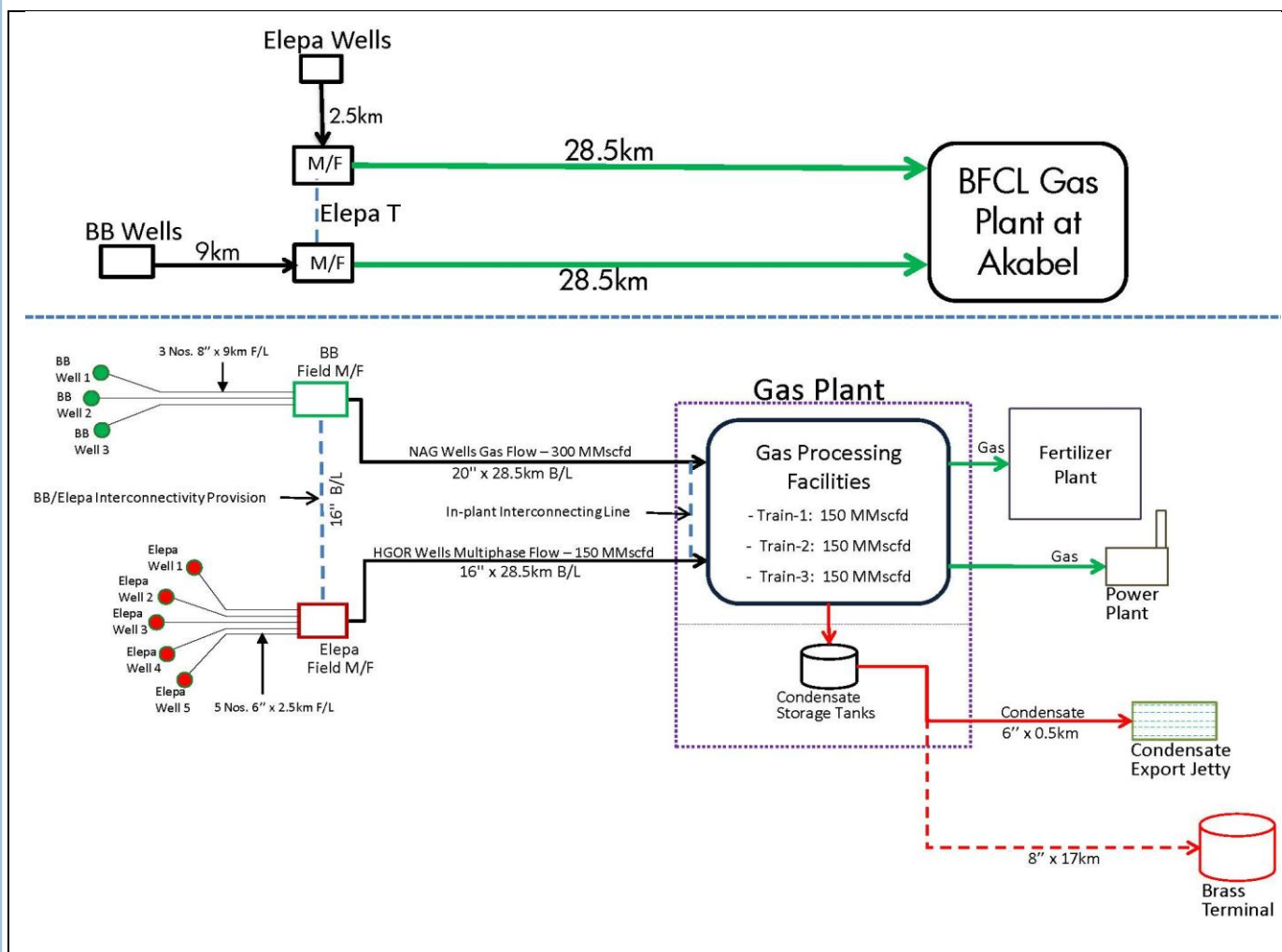




OML 33 GAS SUPPLY TO BRASS FERTILIZER PLANT

PROJECT EXECUTION READINESS ENGINEERING REQUIREMENTS



23 December 2013

SUMMARY PAGE

Client:	BRASS FERTILIZER COMPANY LTD		
Project:	OML 33 GAS SUPPLY TO BRASS FERTILIZER PLANT	Title:	Project Execution Readiness: Engineering Requirements
Facilities:	Gas Gathering Pipelines & Processing Plant	BFCL Project Ref:	EA/BFCL/GEN-001

This document serves as a reference that demonstrates the state of readiness on the engineering activities in respect of gathering and processing OML 33 gas for supply as feedstock to Brass Fertiliser Plant, and a framework for its implementation.

It highlights the project team's understanding of the workscope, the high level engineering & related discipline deliverables, business process, work methods & procedures, quality assurance processes and integration of HSE in engineering design and CAS in execution planning, - which are all integral parts of the readiness.

The document also covers fair details of the engineering works, and the closely related survey & EIA activities to come and how they will be carried out, - as testimonies to the project's execution readiness.

KEY WORDS: BRASS, DESIGN, ENGINEERING, EXECUTION, FERTILIZER, GAS, PLANT, READINESS.

REFERENCES		HOLDS	
Originator	Document No	Description	Page
EA	EA/BFCL/GEN-001		

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Abbreviations

The Following definitions and abbreviations have been used in this document.

Organisational

BFCL	Brass Fertilizer Company Limited
BYSMENV	Bayelsa State Ministry of Environment
DPR	Department of Petroleum Resources
EA	Epic Atlantic Limited
FEPA	Federal Environmental Protection Agency
FMENV	Federal Ministry of Environment
LGA	Local Government Area
NGO	Non Governmental Organisation
SPDC	The Shell Petroleum Development Company of Nigeria Limited

Technical Terms

BB	Bubuwei Bou
B/L	Bulkline
BfD	Basis for Design
CAS	Community Affairs, and Security
CPT	Cone Penetrometer Testing, Cone Penetration Test
EDMS	Electronic Data Management System
EIA	Environmental Impact Assessment
EMP	Environmental Management Plan
EPC	Engineering, Procurement, and Construction
FEED	Front End Engineering Design
FID	Final Investment Decision
FDP	Field Development Plan
FP	Full Preparatory
GPS	Global Positioning System
HAZID	Hazard Identification
HRA	Health Risk Assessment
HSE	Health Safety and Environment
IFC	Issued for Construction
IFD	Issued for Design
IFR	Issued for Review
ISO	International Standard Organisation, Isometric Drawing
MTO	Material Take Off
OML	Oil Mining Lease
OPL	Oil Pipeline License
PFD	Process Flow Diagram
P&ID	Piping & Instrumentation Diagram
P.I.	Point of Intersection
PI	Preliminary Investigation

Technical Terms – contd.

POS	Project Overview Statement
PTS	Permit To Survey
QA/QC	Quality Assurance / Quality Control
QC	Quality Control
QCed	Quality Checked
QMS	Quality Management System
RFQ	Request for Quotation
RoW	Right of Way
SPT	Standard Penetrometer Testing
T	Pipeline Tie-in Point
WHIS	Worksite Hazard Information System

Units of Measurements

Ha	Hectare
km	Kilometre
m	Metre
M	Thousand
MM	Million
MMscfd	Million Standard Cubic Feet per Day
MTPD	Metric Tonnes Per Day
km	Kilometre
m	Metre
M	Thousand
MM	Million
MMscfd	Million Standard Cubic Feet per Day
MTPD	Metric Tonnes Per Day

1. Background and Introduction

Brass Fertilizer Company Nigeria Limited (BFCL) intends to establish a fertilizer plant with capacity of 5,000 MTPD of Methanol and 2,200 MTPD of Ammonia to produce 3,850 MTPD of Urea with a baseline natural gas consumption of 200 MMscfd. The plant will be located in Brass Island, Bayelsa state of Nigeria, near the shores of Atlantic Ocean. This project is expected to come into production in 2016 and to achieve that, FID is planned for June, 2014. In pursuit of this business objectives, BFCL is at advanced discussions/negotiations with Shell Nigeria to off-take Shell's well head natural gas in OML 33, transport and process same in BFCL's facilities in order to avail BFCL the required 200-300 MMscfd feedstock gas. Part of this gas will be used to generate power on which both the gas processing and fertilizer plants will run.

BFCL have had discussions with major Ammonia/Urea/Methanol plant construction companies and shortlisted three EPC contractors, - Hyundai Heavy Industries / Technipe, Saipem SA, and Wuhuan Engineering Company/Tecnicas Reunidas as potential EPC contractors who will progress the fertilizer plant construction works on a turnkey contract basis. A key enabler facility to the fertilizer plant is the natural gas processing plant and associated pipelines which will transport and process well head gas from SPDC's OML 33 for delivery to the fertilizer plant at Brass Island.

This document focuses on the Engineering Design Requirement, which forms part of the Execution Readiness of the Gas Processing Plant. Henceforth, references to the term "Project" will be the Natural Gas Processing Plant & its associated pipelines, and will not include the fertilizer producing plant, or the sister project – the power generating plant. This Gas Plant's main objective will be to supply the other two plants with hydrocarbon and water dew-pointed natural gas. The power generating plant will in turn supply electric power to the fertilizer and the natural gas processing plants.

1.1. Objectives

- Identify project survey and engineering design requirements necessary to progress the project from Pre-Feasibility/Initiation to FID.
- Outline survey processes and procedures required for optimal execution of the work – topographic, hydrographic, and geotechnical.
- Outline engineering processes and procedures required for optimal execution of the work.

- Identify the roles, responsibilities and high level deliverables of each discipline engineering team at each project phase from feasibility study to front end engineering with sufficiently robust definition to enable letting an EPC contract.
- Provide guidelines on normal expected timing for each phase of the engineering work from Feasibility to limited detailed design.
- Outline the HSE requirements necessary to be incorporated into all stages and aspects of the engineering design aspects of the project.
- Introduce risk management as a necessary tool to be deployed in all aspects of the work included in the engineering design phase.

1.2. Work Phases

The discipline engineering design work will apply for the various work packages necessary to enable the gas plant to be designed, procured, constructed/fabricated, installed and commissioned.

The following work phases will be applicable:

- Feasibility
- Concept Selection
- Project Specification / Design Basis
 - Field Development Planning
 - Front End Engineering Design
 - Initial location surveys (geotechnical surveys, topographical surveys, hydrographical surveys, site acquisition processes)

1.3. Project Critical Success Factors

The following are considered the critical success factors guiding the project, and the project will need to be delivered within the boundaries specified by these parameters to be adjudged a success.

- Project delivered on time and this means that the gas processing plant will be commissioned within 27 - 30 months of FID.
- The project approved budget is not exceeded by more than 10%.
- The project HSE objectives which will be fully defined in the project HSE case are fully met.
- Plant availability numbers defined at the project specification documentation will be met and sustained over the project's life cycle.
- Product quality will remain within acceptable limits agreed with the end user 99.5% of the time.

2. Engineering Design Requirements

2.1. Introduction

Engineering design develops the technical understanding of the project and enables progression from challenges, through ideas for solution and concepts, and specifying the necessary hardware and procedures which can perform work to meet project objectives, as desired by the end user or owner. The objective of this section is to elaborate on the systematic process with minimal risks that will be followed in engineering design to arrive at the end product being the natural gas processing plant.

2.2. Engineering Design Scope Definition

Engineering design scope establishes and defines the boundaries and confines of the challenges to be addressed. It is highly essential that the engineering design scope be properly and adequately defined in such a way that the probability of achieving a robust end solution will be increased, while limiting the risk of expending funds on challenges that need not be addressed by the owner. Engineering design scope flows directly from the project scope as it defines the mechanism to address the length and breadth of the project technical issues. Though project scope definition is a predominantly project management activity, it significantly affects discipline engineering work activities and drives their output. Inadequate project scoping ultimately results in a non-robust engineering design which becomes more costly to fix as the project progresses. If not remedied, this may result in a finished product that does not adequately meet all the needs for which it was conceived. Fortunately, Project Management processes are in place to address proper project scoping, and if properly applied at the onset, significantly improve the chances of project success. For completeness, some of these processes are briefly outlined here:

- Project Overview Statement (POS)
- Project Scoping Workshops and Challenge Sessions
- Needs Assessment Sessions

2.3. Adequate Front End Loading

Front End Loading, - a term given to a group of specialist engineering and technical work activities that enable better definition of the project, will be embraced in prescribing the process by which project technical objectives will be met. It includes examining the scope to ensure that it encompasses all the relevant areas that need addressing. It defines what, how and when a project will be accomplished. Because it is multidisciplinary and iterative in nature, heavily reliant on highly experienced technical resources, generating mainly reports, drawings and other documents, but with very little brick and mortar output, it is sometimes thought of as being of little additional value or taking up precious time that

could have been used to commence “fabrication of things”. If not adequately carried out, projects tend to take longer, consume more funds, involve a lot more rework, and tend to less likely meet the full objectives.

2.4. Engineering Document Requirements

In the course of engineering design, key documents that define what and how actions need to be carried out in order to ultimately arrive at the desired end product will be produced. These documents enable a smooth progression from one stage of the project to another. A document management process (Document Control Management) will be put in place to ensure that only the approved versions of documents are carried forward, used and referenced. These documents are the outputs of intellectual engineering work and represent what was agreed and approved to be done, as well as what was done. Engineering documents on the project will come in various forms including reports, calculations, sketches, diagrams, drawings, pictures, videos and software programs and all will need to be stored and retrieved in a professional manner, and this will be the basis of an EDMS which will be deployed on the project.

The following key engineering documentation will be produced in the course of engineering activities on the project.

- Feasibility Study Report
- Concept Selection Report
- FDP Report
- Basis for Design
- Conceptual Design Report
- Project Specifications
- FEED Package
- Detailed Engineering Package
- As-built Documentation Package

The first three on the list will involve other expertise besides facilities discipline engineering resources, and are frequently driven by those other experts outside facilities engineering design team.

The other six document packages are driven by Engineering Design teams and further details on them are provided below:

2.4.1. Basis for Design

Basis for Design (also called Basis of Design, Design Basis Memorandum) is a critical starting document which defines to the design team that which the owner needs to design. It establishes the fundamentals on which the design will be based, project expectations

and the needs which the end product will meet. Basis for Design for this project will address the following key areas:

- Introduction to the Project
- Engineering Scope - Process, Mechanical, Civil/Structural, Electrical, I&C, Telecommunications, and Technical HSE
- Project Deliverables
- Project Success Factors
- Design Life
- Basic Design Parameters
- Existing Utilities Summary
- Instrumentation and Control Philosophy
- Safety and Shutdown Systems
- Design Codes, Standards & Specifications
- Any special issues on the project
- Key Project Team Contacts
- Drawings
- High Level Schedule (for the design, and for the project)

The following checklist will be referenced, as it is relevant to the project:

Overview of field and development	Wellhead Data
Field Location	Export Product Specification to Fertilizer Plant
Units and Co-ordinates	Pipeline Entry Requirement
Wind Data	Pipeline Route / Flowline Design Data
Air Temperatures	Facilities Design Capacities
Air Humidity	Facilities Overview
Soils Data	Operations and Maintenance Strategy
Reservoir Data and Management Strategy	Manning
Produced Gas Analysis Data	Operational Control
Produced Crude Analysis Data	Communications
Produced Water Analysis	Accommodation
Sand Production	Production and System Reliability / Availability
Production Monitoring Methodology	Spares Philosophy
Well numbers and types	Maintenance Philosophy
Drilling and Completion Strategy	Emergency Escape/Safety
Drilling Rig Parameters	Permits and Consents
Well Intervention Strategy and Forecast	HSE Codes / Standards
Production Profiles	Emissions Limits
Production Chemistry	

2.4.2. Conceptual Design Package

- Design Basis Memorandum
- Site Visit Reports
- Site/Route Survey Reports and Drawings
- Preliminary (Conceptual) Design Package
- Master Calculation Sheet
- Document Delivery Register
- Master Equipment List
- Process Control Philosophy
- Loss of Containment Detection, Surveillance, Security and Cathodic Protection
- RFQs
- Project Drawings – Utility Drawings, PFDs, P&IDs, GA's, Isometrics (as applicable)
- Hazop Report
- Post Hazop Drawings (as applicable)
- Project Schedule
- Project Cost Estimate

2.4.3. Project Specifications

- Design Basis Memorandum
- Site Visit Reports
- Site/Route Survey Reports and drawings
- Preliminary (Conceptual) Design Package
- Master Calculation Sheet
- Document Delivery Register
- Master Equipment List
- Process Control Philosophy
- Loss of Containment Detection, Surveillance, Security and Cathodic Protection
- RFQs
- Project Drawings – Utility Drawings, PFDs, P&IDs, GA's, Isometrics (as applicable)
- Hazop Report
- Post Hazop Drawings (as applicable)
- Project Schedule
- Project Cost Estimate

2.4.4. FEED Package

The FEED package when delivered will cover the following listed documents among others. In many cases, on completion of FEED, the project is considered sufficiently defined and detailed to enable a final investment decision to be taken. At this stage the technical risks and uncertainties are better understood and adequate mitigation measures that re-assure the project owners of higher probability of success have been put in place.

It is at this stage after FID that commitments for procurement of long lead items are made and EPC contracts finalized and signed.

- Design Basis Memorandum
- Site Visit Reports
- Site/Route Survey Reports and Drawings
- Preliminary (Conceptual) Design Package
- Master Calculation Sheet
- Document Delivery Register
- Master Equipment List
- Process Control Philosophy
- Loss of Containment Detection, Surveillance, Security and Cathodic Protection
- RFQs
- Project Drawings – Utility Drawings, PFDs, P&IDs, GA's, Isometrics (as applicable)
- Hazop Report
- Post Hazop Drawings (as applicable)
- Project Schedule
- Project Cost Estimate
- QA /QC Plan
- Operations Philosophy
- Control Philosophy
- Contracting Strategy
- Risk Management Plan
- HSE
- Baseline Environmental Plan
- Cathodic Protection Philosophy
- Corrosion Control

2.4.5. Detailed Design Package

- Updated Design Basis Memorandum
- Design Limiting Case
- Design Calculation Notes
- Corrosion Protection Design Basis
- Site/Route Selection Report
- QA/QC Plan
- Operations Philosophy
- Control Philosophy
- Design Safety and Safeguarding System
- Drawings – PFDs, P&IDs, Gas, ISOs (Approved for Construction)
- 3-D Models for the Plant

- Technical Datasheets for all Equipment
- Project Specifications
- All Ancillary Specifications
- Project Security Philosophy
- Civil / Structural
- Structural Steel
- Mechanical and Materials
- Pipeline Mechanical
- Master Document Register
- Electrical
- Instrumentation / SCADA
- Pipeline
- Health , Safety and Environment
- Project Security Philosophy
- Pre-Commissioning and Commissioning
- Master Project Schedule

2.4.6. As-Built Package

- Updated Equipment List
- Post Construction Project Drawings – Equipment Layout, Plot Plan, Overall Piping General Arrangement redlined and back-drafted.
- Site Visit Report
- As-Installed Photos
- Installation Videos
- As-Built Activity Schedule

3. Geomatics Requirements & Deliverables

3.1. Introduction

The surface facility conception studies for the engineering, installation and commissioning of the gas plant and the associated pipelines to transport the well-products from both BB and Elepa Wellheads to BFCL Base in Akabel requires a robust and carefully detail survey processes. This pipeline route is estimated to be 9 km from BB to Elepa-T and 28.5 km from Elepa-T to Akabel

3.2. Survey Objectives

The objectives of this pipeline route surveys are primarily to:

- (a) Select, define and delineate a pipeline route, including depicting all topographical and hydrographical details within and immediate neighbourhood of the entire Right of Way, sufficient enough for use, and to inform the authorities concerned of intentions and obtain their approval.
- (b) Provide survey supporting documentation and maps to enable the application, and consequent approval from regulatory authorities, in respect of the installation and operation of the pipelines and the gas processing facilities.
- (c) Provide survey details to support and assist in the assessment, compensation and acquisition/renting of the desired land for the purpose of installing and operating the pipeline facility.
- (d) Provide precise spatial information to support and form reliable basis for the engineering design, tendering, construction, commissioning and operation of the pipeline.
- (e) Provide relevant pipeline survey documentations to meet the requirements of BFCL's Oil Pipeline License and satisfy the Laws of the Federation of Nigeria, Chapter 338 - Oil Pipelines Act of 1995.

The geomatics requirement for the project execution readiness, is therefore summarized into:-

- Pipeline Route Study and Selection.
- Conduct of Preliminary Investigation (PI) Survey.
- Conduct of Full Preparatory (FP) Survey.
- Land Acquisition and Parcelisation Surveys.

3.3. Pipeline Route Study and Selection (Desktop and Field Inspection)

Pipeline route selection, definition and refinement, as an iterative process, involves the initial desk-top studies to identify routing options, taking into account known key environmental, cultural, geological, archaeological, geotechnical and hydrological sensitivities. The result of the study will enable the development of basic pipeline routing information from available topographical and geological maps, aerial photography and/or satellite imagery, and available literature, including consultations with the relevant land owners and stakeholders and finally a review of the EIA criteria.

The ultimate aim is to achieve an economically and environmentally feasible route for construction and operation of the pipeline.

3.3.1. Key Routing Principle and Influencing Factors

The key principles to take into account when performing route selection are:

1. Safety of the public and personnel - the route must provide a safe and secure environment for the pipeline during construction and over its operational life and ideally be routed away from populated areas.
2. Economics: - the route should meet the project's economic objectives, without compromising safety and environment and minimizing local economy impact on communities that the pipeline passes through, and have the smallest footprint feasible (ideally the shortest distance between pipeline start and end points).
3. Land ownership related factors e.g. the number of landowners, anticipated ease and cost to obtain/purchase consents.
4. Easement width.
5. Contents of the pipeline and operating conditions, e.g. consideration of leakage of a high vapour pressure liquids.
6. Environmental impact: – the route must have a minimum negative impact on the environment and minimum land use.
7. Terrain and subterranean conditions, including geotechnical, hydrographical, and meteorological conditions. This includes ground stability, including other land uses which may create instability (e.g. mine subsidence, land development/excavation).
8. Cultural heritage sites.
9. Existing and future land plan usage. This can be determined by research of public records and consultation with land planning agencies which should identify:
 - third-party activities
 - agricultural practice
 - existing facilities and services
 - future developments

10. Permanent access: – the pipeline must ideally be accessible for inspection and maintenance all year round over its operational life.
11. Security: - The pipeline system should be routed to minimise security concerns, particularly due to trespass and sabotage, during both construction and operation.
12. Other unforeseen hazards.

3.4. Preliminary Investigation (PI) Survey

The preliminary investigation survey shall be conducted for site evaluation of the viability of the selected route from the desktop studies. The most preferred of the proposed alternative routes is investigated on the ground by a quick survey process. The survey route shall follow as close as possible one boundary of the proposed way leave. Bush clearing of survey route shall be made to cause minimum damage to existing natural features like large trees or similar features. Where significant obstructions to construction convenience are discovered, their extent is determined and the possibility of deviating from the route should be investigated. Survey pillars shall be placed at all PI points, while stout wooden pegs shall be used as traverse points.

The river and creek crossing points shall be identified and investigated for construction convenience and other potential hazards and constraints. Other critical factors for consideration during the PI Survey includes:-

1. Significant elevation difference at the overland area around the rivers/creeks crossing points.
2. Height of adjacent bank. Abrupt changes of elevation (i.e. steep banks) to be avoided.
3. The approaches to the crossings shall be direct (i.e. without sudden alteration of direction). Preferable, the approach should be close to perpendicular to the river/creek course.
4. The crossing length should to be as short as possible.

3.4.1. Survey Methodology and Resources

The survey methodology shall be a conventional topographical and hydrographical survey method using acceptable instrumentation processes. All data acquisition shall be by automatic data capture and logging. The field survey team will typically consist of Geomatics Engineers, Field Surveyors, Survey Assistance and Labour. Support personnel shall include drivers (vehicle and boat), nurse, community relations, and security personnel. While office support shall include the data processing, quality assurance and charting team, using approved software.

3.4.2. Survey Deliverables

At the end of field survey, the acquired data shall be processed and QCed for compliance with agreed technical standard and specification.

Twelve (12) copies of typically formatted route map suitable for PTS application, on a scale of 1:50,000 and adequately annotated, shall be produced. This is part of the supporting documents for PTS application.

As a minimum the map shall indicate Start and End points of proposed route, with spatial description of pipeline route and estimated length of pipeline route (km). These drawing will form part of the attachments for PTS application.

Please note that with a robust desk top studies and analysis based on available information and combined with site visits to further investigate the physical nature of the selected route and consultation with the local communities on other identifiable constraints, the physical survey requirement in the phase could be avoided.

3.5. Full Preparatory (FP) Survey

With the granting of the Permit To Survey (PTS), a Full Preparatory Survey shall be conducted along the selected route using very robust survey methods and processes. The survey will follow both boundaries of the eventual right of way of the pipeline with composite changes of direction at the selected Points of Intersection. During the survey, all topographical and spatial details of existing natural and man-made features, including marine, terrain and vegetation description, within the RoW and 50m outside of the RoW limits shall be detailed. All buried features and facilities shall be detailed, including precise determination of burial depths. The edges of the RoW shall be cut open, marked and delineated with suitable survey monuments at all intersection points and at every 200 metres along the survey lines, except otherwise specified.

3.5.1. Hydrographic Survey

Hydrographic survey shall be carried out at all the river and creek crossings, along the axis of the RoW and extending to at least 200m on either sides of the limits. The bathymetric sounding shall be reduced to the same elevation datum as the main pipeline topographical survey. Tide gauges shall be installed at strategic locations for optimal coverage of the bathymetric survey, for use in referencing and reducing the soundings. Tide gauges elevation shall be tied to the topographical survey control.

3.5.2. Survey Methodology and Resources

The full preparatory survey shall be primarily based on Global Positioning System (GPS) and tied to National Grid Co-ordinate System using a 7 datum shift transformation parameters.

The survey methodology shall be a conventional topographical and hydrographical survey method with an acceptable instrumentation, data process and quality assurance softwares. The field survey team will typically consist of Geomatics Engineers, Hydrographic Engineers, Navigators, Field Surveyors, Survey Assistance and Labour. Support personnel shall include drivers (vehicle and boat), nurse, community relations, and security personnel.

Summary of the FP survey processes includes but not limited to:

- GPS Establishment and Control Densification
- Topographical Survey with feature detailing.
- Hydrographical Survey.
- Data processing and Quality Assurance
- Charting.

3.5.3. Survey Deliverables

At the end of field survey, the acquired data shall be processed and QCed for compliance with agreed technical standard and specification. A complete set of data relevant to engineering design and construction, including safe and reliable operation processes of the pipeline shall be produced.

Key Deliverables include:

1. Typical OPL formatted pipeline route map at the scale of 1:50,000, indicating points of intersection (P.I.), including bearings and distances of the lines. Sufficient copies of this map shall be produced and used as one of the attachment for the OPL application
2. Alignment sheet templates at an appropriate scale, indicating plan and profiles of the route at a scale of 1:2500. Each sheet shall cover not more than 2 km section of the RoW with clear match-line of adjacent sheets.
3. Dedicated Major River crossing chat, indicating plan and profile of the hydrography and adjoining topography.

3.5.4. River Crossing Chart Details

The dedicated major river crossing drawing shall be in two parts - plan and a profile, and showing all details in the vicinity and approaches of the crossing.

Scale 1:500 (horizontal), 1:50 (Vertical), or for longer crossings

1:2,500 (horizontal), 1:50 (Vertical)

The plan shall show all relevant features including:

1. The bank limits of the river channel.
2. Reduced elevations (spot heights) of the surveyed marine and adjoin land sections.
3. Direction and rate of water flow.
4. Description of nature of the creek bed.
5. Direction and distance to adjacent P.I. points.
6. The tidal characteristics with indication of the expected highest and lowest tide levels.
7. Adjoining boundary survey pillars with details on either bank.

3.6. Pipeline Route Land Acquisition and Parcelisation Survey

The survey support requirement during the land acquisition of the pipeline route is summarized as follows:

1. Prepare and issue an Area Advice, indicating the total computed area to be acquired in Hectares.
 - With the land acquisition team, conduct site verification and assessment with the land owners and delineate limits of individual land ownership.
 - Process and prepare a parcelisation map showing individual land ownership limits within the entire route. Each parcelated section shall contain name of the original owner and the corresponding area acquired in Ha.

3.7. Gas Plant Site Full Preparatory (FP) Survey

The gas plant area is intended to be part of and within BFCL complex in Akabel.

The full preparatory survey is therefore to provide precise spatial information to support and form reliable basis for the front end engineering design, general layout and hydrological design, tendering and construction.

3.7.1. Gas Plant Full Preparatory Survey Process

The full preparatory survey shall be primarily based on Global Positioning System (GPS) and tied to National Grid Co-ordinate System using a 7 datum shift transformation parameters. This survey shall also be tied to the pipeline route survey control system.

The boundary limits shall be delineated and marked with standard survey monuments, while the topographical survey shall be on 50 x 50 m grid.

The field survey team will typically consist of a Field Survey Lead, Survey Assistance and Labour. Support personnel shall include drivers (boat), nurse, community relations and security personnel.

Summary of the FP survey processes includes but not limited to:-

- GPS Control Establishment and Densification
- Boundary delineation and marking.
- Perimeter survey.
- Topographical Survey.
- Data processing and Quality Assurance
- Charting.

3.7.2. Survey Deliverable

At the end of field survey, the acquired data shall be processed and QCed for compliance with agreed technical standard and specification. A complete set of spatial data relevant to engineering design and construction shall be produced.

Key Deliverables include:-

- GPS survey report with station descriptions.
- Topographic survey drawing with contour lines, existing features, vegetation and terrain descriptions.

3.8. Geotechnical Surveys

The soil investigation survey is required to determine the prevailing nature, type and characteristics of the sub soil layer within the Right of Way, including major river and pipeline crossings. It will also enable the gathering of sufficient site specific geotechnical data for front end engineering design (FEED) consideration, installation of the pipeline and geo-hazard studies and analysis of the route. The survey shall be a shallow geotechnical investigation type, which will incorporate cone penetration testing, in-situ test and laboratory analysis.

3.8.1. Pipeline Geotechnical Survey Process

Generally, the soil investigation survey shall be carried out as follows:

- Mobilize and demobilize field equipment and personnel to-and-from site for sample acquisition and in-situ tests.
- Carry out percussion drilling of boreholes at predetermined 3 km intervals to a minimum depth of 4 m, by means of a manual rig equipped SPT accessories (or alternatively with an hand auger). Including one (1) vane testing for direct shear strength (friction angle and cohesion) to 5 m or to point of refusal adjacent to the auger locations.

- Execute Standard Penetrometer Testing (SPT) every 1.5 m advance into the bore holes. Also carry out Cone Penetrometer Testing (CPT) to maximum termination depth or point of virtual refusal, whichever occurs first during the course of drilling. The discontinuous mechanical CPT method is the preferred option.
- Collect and preserve disturbed and undisturbed samples from non-cohesive and cohesive soil from the bore holes for laboratory analysis.
- Carry out visual identification and field classification of recovered samples.
- Carry out electrical soil resistivity measurements at every drill point to determine the corrosivity of the environment where the pipelines will be laid.
- Identify and locate all geohazards within the area of interest and execute the necessary in-situ tests to quantify the associated risks.
- Additional boreholes shall be drilled on either bank of the major river crossings to a depth of 10 m. Each borehole drilling activities shall be accompanied by SPT tests. Test intervals shall be one (1) test per one (1) metre drilling depth.
- For minor creek crossings more than 150 m, one (1) borehole to depth of 10 m plus one (1) CPT, on one bank of the creek, shall be carried out.
- All data processing and soil investigation analysis shall be conducted at company base using acquired raw data and soil samples.

3.8.2. Geotechnical Survey Methodology and Resources.

Below is the summary of the geotechnical survey method:

Along Pipeline RoW

- One (1) Percussion rig boring to a depth of 4 m every 3 km along the pipeline route.
- One (1) vane testing for direct shear strength (friction angle and cohesion) to 4 m or to point of refusal adjacent to the auger locations.
- One (1) electrical soil resistivity measurement every 3 km along the pipeline route.

At River Crossings

- For major creek crossings (> 200 m) :- Two (2) boreholes to depth of 10m plus two (2) CPTs, one on either bank, shall be carried out.
- For minor creek crossings 100 - 200 m) :- One (1) borehole to depth of 10m plus one (1) CPT, on one bank, shall be carried out.

3.8.3. Gas Plant Geotechnical Survey Process

Plot size of about 500m x 500m within the BFCL complex in Akabel is envisaged for the Gas Plant. The Geotechnical investigation survey process of the Gas Plant site shall be covered as follows:-

- Drill five (5) Boreholes at the designated coordinates to a depth of 30 m each, obtaining undisturbed and representative samples at appropriate intervals of depth, and performing standard penetration tests at 1.5 m depth intervals.

- Drill two (2) Boreholes at the designated coordinates to a depth of 15 m each, obtaining undisturbed and representative samples at appropriate depths, and performing standard penetration tests at 1.5m intervals of depth.
- Boring of eight (8) Hand Auger holes of a depth of 5m each, obtaining representative samples and logging as appropriate.
- Excavation of eight (8) Investigation Pits to a depth of 2.5m each, obtaining undisturbed and disturbed soil samples and describing as appropriate.
- Performing fourteen (14) Cone Penetration Test (CPT) at designated locations to depths of penetration resistance or depths at which the anchorage capacity of the machine can permit.
- Carrying out eighty-six (86) Soil Resistivity Surveys at specified positions using Wenner or 4 - terminal methods.
- Conducting in-situ tests and laboratory investigations on soil and water samples obtained from the Boreholes, Hand Auger Holes and Investigation Pits, as specified.

3.8.4. Geotechnical Survey Deliverables.

At the end of field works, all acquired samples shall be tested in the laboratory and all results assembled in a concise report. The geotechnical report shall include both factual results and interpretative report section.

The report and associated chart shall contain the following details:

- Executive Summary
- Summary Description of Work Methodology,
- Summary Description of Processed Results.
- Conclusion and Recommendations.
- Charts indicating Spatial Definitions of Soil Test Positions.

4. EIA Requirements & Deliverables

4.1. Introduction

The surface facility conception studies for the engineering, installation and commissioning of the pipeline to transport the well-products from both BB and Elepa Wellheads to BFCL Base in Akabel, the condensate disposal line from Akabel to either Brass or Nembe, and the Gas Plant at Akabel, requires a detailed environmental impact assessment processes at the Gas Plant site and along the pipeline routes and environs.

The EIA addressed here are only in respect of the pipeline routes from BB though Elepa to Akabel. The EIA for the Gas Plant site is already ongoing, while the condensate evacuation routes are options yet to be decided.

The regulatory framework for EIA implementation in Nigeria is the EIA Act No. 86 of 1992. The Law confers the mandate to implement EIA on the Federal Ministry of Environment. In accordance with this law, it makes it mandatory for proponents of all new major development activities to carry out EIA on their proposed project.

The EIA requirement therefore, shall be broken down into, EIA Objectives, EIA Process, EIA Scope of Works and EIA Works Methodology.

4.2. Objectives of the EIA

The objectives of the EIA shall include but not limited to the following:

- Describe the integrated environmental baseline context of the pipeline route (ecological, biophysical, chemical, socio-economics and health characteristics).
- Qualify and Quantify potential impacts (magnitude, duration, frequency, etc) of the project on the integrated environment and consider measures to avoid, mitigate or ameliorate them. The approved FMENV quantification method will be adopted.
- Identify from detailed operations and activities the following: - hazardous materials, waste generation, emissions and discharges, pollution expected from the activities.
- Identify changes in health determinants that may result from the different phases of the project and evaluation of local population exposure to these changes.
- Investigate all applicable national and international regulations relevant to the project.
- Develop feasible measures, procedures and practices to be followed during design, construction and operation to ensure that the environmental effects are mitigated or minimised.
- Develop environmental management plan including compliance monitoring, auditing and contingency planning.

4.3. Requirements of the EIA Process

4.3.1. Registration of Project

An application letter should be written to the Honourable Minister of Environment requesting for an EIA permit to embark on proposed project. This application should be accompanied with the following:

- A duly completed registration form which should be in duplicate
- A bank-draft for the sum of Fifty thousand Naira (N50,000.00) addressed to the Federal Ministry of Environment and
- A project proposal (ToR)

4.3.2. Registration of Project

A Terms of Reference (ToR) for the preparation of the project's draft EIA reports should be prepared and submitted by the proponents (Brass Fertilizer Company Limited). Thereafter, following the approval of the TOR, field data gathering and laboratory analysis will be witnessed by the FMENV's officials, during which process FMENV accredited consultants / laboratories should be used.

4.3.3. Registration of Project

Following the categorization of the project's EIA by the Federal Ministry of Environment, a specified number of copies of the project's draft report will be submitted depending on the category of the project. This is accompanied with the Initial Processing Fee for the proposed project.

4.4. EIA Scope of Work

4.4.1. Detailed Workscope of the EIA Study

The detailed workscope of the EIA shall include but not limited to:

- Reconnaissance Survey of the Project Area.
- Literature Review and Field Observations of existing baseline data, as well as undertaking fieldwork, environmental data acquisition and laboratory analysis to fill information/data gaps.
- Full description of project activities and identification of potential impacts associated with logistics, site preparation, construction, new installations, operation, abandonment and specific risk/hazards at each stage of the project development.
- Qualification of potential impacts, their prediction and evaluation of their significance using appropriate models.
- Identification of effective mitigation measures for the project activities.

- Identification of all communities within the project location for effective consultation, including FMENV, BYSMENV and Brass LGA.
- Determination of social and socio-economic effects of the proposed project on the communities.
- Development of comprehensive Environmental Management Plan, including monitoring, decommissioning/abandonment and remediation plans.
- Remediation Plan after Decommissioning, Closure or Abandonment.
- Writing of Reports that conform to the standards and guidelines set by the Federal Ministry of Environment.

4.4.2. EIA Methodology

The methodology for conducting the EIA shall involve:

1. Literature Review

This will involve the study of existing literature particularly, from reports of previous EIA studies in the area, and other relevant studies on the environmental characteristics of the study area. Materials to be reviewed shall include textbooks, reports, survey maps, aerial photographs, articles and journal papers and any previous EIA reports from the area / or similar projects.

2. Reconnaissance Survey

A reconnaissance survey of the study area shall be undertaken to familiarize the EIA team with the proposed project area. This will help them in the concept design of the field studies execution.

3. EIA Study Consultation Programme

In line with the FMENV 1995 Procedural Guidelines for EIA in Nigeria, there shall be consultation with the host communities, recognized Community Based Organisations and other interested stakeholders on the proposed project area. Their views/ideas/opinions will be integrated into the EIA project as appropriate.

4. Field Work Activities / Laboratory Analysis

Fieldwork activities / laboratory analysis shall be carried out to verify data gathered from literature and to collect additional data to fill information gaps. The activities shall be carried out in accordance with FMENV, APHA, IITA and USEPA methodologies.

Due to the nature of the project, construction of fertilizer plant and facilities site, the construction activities and operational phases that are anticipated to generate some negative impacts. However, there is high beneficial impact, contributing significantly to economic development of the area, state and nation. Also based on available previous environmental studies from closeby Yenagoa and Brass areas, it is strongly recommended

that data acquisition / fieldwork should be for two sampling seasons. This will enable the collection of data to update information in the area.

5. Environmental Impact Assessment

This shall involve impact identification, prediction and evaluation. Impact evaluation shall be carried out using standard approved methods, such as the "Strength of Relationship Matrix Approach" that is applicable, specific and quantifiable. It is a method that defines numerically, the degree of interdependence of the various environmental parameters to be considered. The impact evaluation results shall form the basis for developing impact mitigation measures and the EMP of the proposed project.

6. Report Writing

There shall be a documentation of the EIA findings in a report format as specified in the FMENV Procedural guideline (FEPA, 1995).

4.5. Impact Assessment, Mitigation & Alternative Monitoring Plan

4.5.1. Assessment of Associated and Potential Impacts

The identification, prediction and evaluation of associated and potential impacts shall be carried out with the standard methodologies including:

- Identification of impact sources – checklist
- Identification of impact indicators – checklist
- Prediction of impact magnitude
- Evaluation of importance of environmental components – consensus of opinions
- Evaluation of impacts – "the Strength of Relations Matrix"
- Identification of mitigation measures; avoidance, elimination / minimization and enhancement strategies for beneficial impacts.

The associated and potential impacts of any proposed road and bridge construction activities shall be described for the various phases of the project as follows:

- Impacts resulting from project location/sitting
- Impact resulting from project construction activities
- Impact resulting from project operation activities
- Impact resulting from project abandonment

The impact shall further be classified as:

- Incremental impact

- Adverse/Beneficial impacts
- Cumulative impacts
- Short term or long term impacts
- Temporary or permanent impacts
- Direct or indirect impacts
- Impacts associated with project risks and hazards
- Reversible or Irreversible impacts

Where an impact is uncertain, the uncertainty will be made explicit and in this respect, risk assessment method will be required.

4.5.2. Impact Mitigation Measures

All impacts identified shall be considered for mitigation and control through preventive, mitigation and ameliorative measures. These measures shall be incorporated in the proposed development to minimize or completely eliminate the key impacts. Some of the measures will include:

- Emergency response and waste management procedures
- Pollution abatement/procedures at all times
- Environment awareness programmes for construction and operation staff
- Practical measures for rehabilitation of impaired features after project construction
- Proposal of changes in schedule of associated activities.

Where the effectiveness of mitigation measure is uncertain, or depends on assumptions about operational procedures, monitoring programmes or management procedures will be defined. Otherwise, the following alternatives will be considered:

- No project option
- Site location alternatives.

4.5.3. Monitoring Plan

This shall be defined and carried out to provide specific information on the characteristics and functioning of environmental and societal valuables in space and time.

4.5.4. Environmental Management Plan (EMP)

The EMP shall clearly specify guidelines for ensuring conformance of project implementation with procedure, practices and recommendations outlined in the EIA report.

The guidelines shall as a minimum include the following:

- Guidelines for ensuring conformation of detailed design with concept design
- Guidelines for implementation programme
- Guidelines by which objectives and commitment will be achieved
- Guidelines for responsibilities and accountabilities
- Guidelines for procedures for dealing with changes and modification of project
- Guidelines for corrective action which will be employed should the need arise
- Guidelines for inspection, auditing and monitoring of all phases of the proposed project
- Guidelines for decommissioning and abandonment of the project.

4.5.5. Quality Assurance / Quality Control

1. Field Work

- Guidelines for procedures for dealing with changes and modification of project will be followed.
- Geo-referencing of all the sampling points will be carried out.
- Establishment of a station numbering system and pre-labeling of all containers, etc. before commencement of fieldwork.
- Design and use of appropriate fieldwork recording forms.
- Establishment criteria for accuracy of positioning requirements.
- Duplicate Samples: - one duplicate sample will be taken for every 10 to 15 samples taken. Half of the duplicate samples will be indicated for laboratory use while the other half will be labeled such that the laboratory will not recognize their duplicates.
- Calibration of field equipment: - all field equipment will be calibrated at the beginning of the measurements (initial calibration).
- Regular meetings during fieldwork will be held to assess progress and performance.

2. Water Sampling

- Care will be taken not to disturb the sediment while taking water samples.
- Samples will not be taken at points near platforms or piers, because the unnatural water flow may disturb the representativeness of the sample.
- Sampling will be taken away from the influence of outboard gasoline engine.
- Samples for both water and sediments will be always collected from downstream to upstream.
- When water and sediments are taken from the same area, water samples will be collected first.

3. Soil Sampling

- Composite soil samples will be taken and portions cut from them (Sub Samples).

4. Laboratory Analysis

- Laboratory QC checks cover all practices and activities that provide the accuracy and precision of the analytical measurements. They serve as a tool for the analyst and for the laboratory to approve the correctness of the generated analytical data. The laboratories used will follow all the QC requirements specified by each analytical method.
- Reagent Blank: - analyte-free water analysed with the samples, one per sample set.
- Preparation Blank: - used when the method includes any pre-treatment method (digesting, distillation, etc.), one analyte-free water blank will be processed and analysed the same way as the samples. It will be used to detect any contamination.
- Others are blind QC check samples, duplicate samples and blind field duplicates.

4.5.6. Environmental Management Plan (EMP)

An appropriate plan for the restoration of the environment in case the facility is decommissioned, closed or abandoned.

4.6. Consultation with Regulatory Agencies / Other Stakeholders**4.6.1. General**

Throughout the duration of the project, an effective communication shall be maintained with relevant regulatory agencies at the Federal, State and local Government levels as well as with identified stakeholders on the proposed project.

Parties of interest for consultation shall include the following:

- Federal Ministry of Environment
- Bayelsa State Ministry of Environment
- Brass Local Government Authority
- Community Based Organisation
- Non-Government Organisation

4.6.2. Objectives of Consultation

The objective of this consultation is to:

- Address issues promptly to avoid conflict.
- Building consensus on potential impacts identified and proffer mitigating measures before the project takes off.
- Avoid any misunderstanding about the development.
- Ensure that any apprehension and fears about the project, nature, scale and impact of the operation have been addressed.

4.6.3. Implementation

- A Public forum involving identified stakeholders shall be organized near the project site with opportunities for them to be jointly and severally acquainted with the project.
- All comments and recommendations of all stakeholders shall be incorporated into the EIA report as appropriate.

4.7. Field Work Activities / Laboratory Analysis

Fieldwork activities/Laboratory analysis shall be carried out to verify data gathered from literature and to collect additional data to fill information gaps. The activities shall be carried out in accordance with FMENV, APHA, IITA and USEPA methodologies.

Data acquisition shall be for two seasons sampling period.

4.7.1. Description of Environmental Components

The description of the environment specific to the study area shall be provided in the EIA report. The environmental elements to be described shall include:

Physical Characteristics

Climatology / Meteorology

Carry out a minimum of 6 one hourly sample measurements in both down and upwind direction of the reference area/facility.

- Temperature
- Rainfall pattern
- Sunshine
- Humidity
- Wind speed
- Wind direction
- Seasonal variation and extreme micro climates, and determine factors.

Air Quality / Noise

Also carry out measurement in for ambient air quality.

- Parameters required are CO, CO₂, SO_x, NO_x, N₂O, NH₃, VOC/HC, SPM, soot, dust,
- Measure ambient Noise levels and for vehicles and equipment in use at project site.

Soil / Land Use

- Carry out a survey and provide a full description of the land use pattern of the project area.

- For the proposed ± 40 km length of gas pipeline, soil samples shall be taken every 2km at surface (0-15 cm) and subsurface (15-30 cm) levels, giving a total of 40 soil samples
- Colour code soil types and photograph.
- Identify important agricultural, forestry and nature conservation, historical relic resources/areas.
- Using the grid method, establish soil sampling stations at surface and sub-surface along transects and randomly.
- Carry out particle/grain size analysis of soils.
- Carry out physic-chemical analysis on the soil samples

Aquatic Systems / Hydrodynamics

- Survey local rivers and estuaries and watercourses in the area, to establish flow rates and movement with seasonal or tidal variations (if any).
- Determine current velocity, water dimensions (depth, width); take Samples (sampling strategy: - very important e.g., spring/neap tides) and have them analyzed.
- Establish appropriate geographical positions/co-ordinates of sampling stations with GPS equipment.
- Collect water samples for the determination of suspended particulate concentrations, bed density, in the major and minor river systems around project site.
- Surface water bodies identified from the map of the proposed gas pipeline include 21 creeklets, 8 creeks and 4 major rivers.
- Based on this, surface water samples shall be collected 10, 8 x 2 and 4 x 2 (downstream and upstream of water crossings) stations for the creeklets, creeks and major rivers respectively. This will give a total of 34 stations.
- At least three (3) samples per station (at the top, mid water and bottom of the water body) shall be taken giving a total of 102 water samples.

Identify water usage – transport, domestic, agriculture etc.:

- Surface water system identification and characterization
- Water quality – nutrients, DO, SM, pH, THC/PHC, heavy metals
- Water microbiology
- Economic importance of aquatic ecosystem

Sediment Samples

- Collect sediment samples at each of the surface water sampling stations, giving a total of 34 sediment samples.
- Colour code sediment types and photograph.
- Carry out particle/grain size analysis of sediment samples.
- Carry out physic-chemical analysis on the sediment samples.

Hydrogeology

- Describe for area in terms of the physiographic and tectonic setting and geological history of the region, superficial and engineering geology with particular reference to properties that are pertinent to review of the project including: Bathymetry and microtopographic of the area;

Biological Characterisation**Biota**

- Using the most current scientific method, carry out sediment, benthos, microbiology, fisheries and pelagic plankton, epipelagic, encrusted species sampling (Intertidal/coastal communities, where applicable) at each of the surface water sampling stations.
- Fisheries sampling should take into consideration along with the water hydrology, fish type, species, spawning grounds, breeding habits, migration pattern and route, catch per unit effort in the area. Identify indigenous fishing methods/gears.

Wildlife / Biodiversity

- Establish, by observation, trapping, reference to specialist institutions and experts and otherwise, the abundance and distribution of wildlife in the following groups; Amphibians, reptiles, fish, mammals, birds, plants, plankton, Endangered, endemic species etc.
- Key communities or dominant species (give scientific, common and local names). Identify wildlife conservation areas, buffer zones and distances from communities.
- For Mammals: include population size, seasonal distribution and movements onshore, near shore and terrestrial species: designation of areas important to any species e.g nursery, feeding, calving.
- For Bird: seasonal distribution, movements and abundance of marine and other bird populations for onshore, near shore area: location and population estimates of sea bird colonies; environmental features affecting the timing of nesting and breeding and over-wintering areas of water fowl.
- Sensitive species: The identification of any species which may be sensitive to the proposed development, and that act as important food resources for other co-habiting species.
- Historic trends in use of the areas by animal populations including those of direct and indirect importance as well as those endangered by man's activities.
- Enumeration should include habitat types and species assemblage (aquatic, intertidal, Swamp Groups etc). Delineate sensitive habitats for these-sea grass habitats, algal beds, hard bottom areas etc).

Vegetation

- Map vegetation types and provide a land capability classification:
- Types of forest, farmlands, plant taxonomy/pathology tissue analysis herbarium studies; classify as undisturbed (forest raphia/swamp); or disturbed (farm bush: natural plant community or plantations, man-made community) or human settlement.
- Provide a map showing species location within its ecological niche and proximity with area.
- Establish, by observation, satellite imageries and consultation with experts and otherwise, the abundance and distribution of vegetation.
- Show the kind of habitat sampled. Identify lichens and other non-conventional vegetation flora.

Socio-Economics

- Generate data using questionnaires and/or statistics from institutions/government agencies on the communities involved (about 6 major) based on the following parameters; economic (occupation/commerce), demographic, housing (settlement pattern), recreational / tourist areas, infrastructure (road, sewage systems, leisure/recreation provision), education, health facilities, income distribution, archaeological / historic cultural resources & general quality of life, shrines, sacred forests.
- Stakeholder's perception (CBOs, NGOs and other identified interest groups within area), attitude and behavior towards the proposed project to be covered in this study. Take into consideration age, sex, status etc.

Community Health Survey

- Structured symptom questionnaire (health and personal history, socio-cultural, existing health symptoms) oral interview, physical/ clinical observation; objective evaluation of eyes, respiratory function, ears, skin, oral hygiene.
- Local community health statistics acquisition; disease trends, existing health care provisions/services (distribution, nos., type, quality) population size served, water supply, sanitation facilities. Obtain information on birth rates/ death rates.

Waste Inventory

- Identify various sources of wastes around project site. Identify sources of wastes and local disposal methods of wastes.
- Determine the ultimate destinations of wastes. Identify local waste recycling techniques.

4.8. EIA Deliverables

The deliverables shall be reports written/produced in accordance with the FMENV Standard. It shall be produced and submitted to Brass Fertilizer Company Limited in phases as indicated below:

- **Terms of Reference (TOR), Reconnaissance Survey Report**

- **Field Report**

This will fill data gaps identified during desktop/literature review and activities carried out during the fieldwork. It will include safety/quality assurance methods, sampling methods, sample preservation methods, data of in situ analysis and various equipment used.

- **Initial Baseline Report**

- **Baseline Report and Draft EIA Report**

This will provide detailed information on desktop study, literature reviews, field data and laboratory analysis of the sampling period. It will also contain preferred options, impact identification, prediction and assessment, mitigation measures, environmental management plan and recommendations.

- **Final EIA Report**

The Final Report shall be submitted within four (4) weeks after completion of the Panel Review and receipt of comments from FMENV. The Final Report will address all comments/problems raised by the regulatory authorities and other stakeholders during the duration of the studies, in addition to other items raised in the draft report.

4.9. Conclusion

A detailed description of the Scope of work for the Environmental Impact Assessment (EIA) study of the proposed BFCL pipeline route project from Babuowei to Akabel (near Brass) in Bayelsa State has been covered in this Terms of Reference (ToR). The EIA workscope considered the existing laws, guidelines and standards on which the project shall be carried out. The key result of the EIA will be used as a basis of communication to obtain relevant approvals and to obtain productive interaction within the public and also serve to ensure environmental sustainability of the project.

4.9.1. Resources

The resources includes very high profile professionals, consultants and specialists, including Marine Biologist, Hydro-biologist, Hydro-geologists, Soil Scientists, Social Scientists, Environmental Economists, Microbiologist, Wildlife, Aquatic and birds Experts. The associated equipment and tools are varied and of state of the art standard.

5. Engineering Disciplines Deliverables

5.1. Introduction

This section will detail the deliverables expected from the various engineering disciplines from Basis for Design stage to FEED and detailed design to such a depth that construction and fabrication can be carried out. They are arranged according to each discipline for ease of reference.

5.2. Process and Systems Engineering

5.2.1. Definitions, Roles and Responsibilities

Process Engineering is the first point of call in engineering design for process equipment in the oil and gas industry. It is an indispensable front end engineering design expertise and signifies the beginning of engineering design for oil and gas process equipment design. It attempts to determine the technical feasibility of an oil and gas operations and provide the mechanism by which a chemical or physical process occurs. The process engineering experts carry out process simulation, heat and material balance and other chemical engineering computations using commercially available specialist software.

5.2.2. Deliverables List for Process and Systems Engineering Discipline

- Process simulations report
- Pipeline sizing report
- Pipeline hydraulics design
- Heat & materials balance report
- Legend and symbols - piping/mechanical
- PFD - inlet facilities
- PFD - heat exchangers, lt separator & sales gas system
- PFD - condensate stabilization
- PFD - overhead gas compression system
- PFD - glycol regeneration system
- PFD - fuel gas system
- PFD - heat medium system
- P&ID - inlet facilities test separator
- P&ID - inlet facilities - inlet separator
- P&ID- inlet facilities - inlet valve skid
- P&ID- gas/gas heat exchanger
- P&ID- gas - liquid heat exchanger / LTS
- P&ID- sales gas metering system
- P&ID- condensate stabilization - LTS liquids flash separator
- P&ID- condensate stabilization - stabilizer feed drum
- P&ID- condensate stabilization - stabilizer feed bottom exchanger

- P&ID - condensate stabilization - condensate stabilizer
- P&ID- condensate stabilization - stabilizer bottoms pumps
- P&ID - recycle gas compression system - 1st stage suction scrubber
- P&ID - recycle gas compression system - 1st stage compression
- P&ID - recycle gas compression system - 2nd stage suction scrubber
- P&ID - recycle gas compression system - 2nd stage suction and discharge bottle
- P&ID – recycle gas compression system - 3rd stage suction scrubber
- P&ID - recycle gas compression system - 3rd stage suction and discharge bottle
- P&ID - booster compressor - suction scrubber
- P&ID - booster compressor - suction blowcase
- P&ID - booster compressor
- P&ID- HP relief valve header
- P&ID- LP relief valve header
- P&ID- HP flare knockout drum
- P&ID- LP flare knockout drum
- P&ID- HP flare stack
- P&ID- LP flare stack
- P&ID- fire water protection system
- Utilities - fuel gas supply header
- Utilities - fuel gas heater
- Utilities - fuel gas knock out drum/distribution
- P&ID- closed drain system
- Utilities - heat medium heaters
- Utilities- heat medium expansion tank and filters
- Utilities- heat medium pumps
- Utilities- heat medium cooler
- Utilities- heat medium heater
- Utilities- compressor lubrication
- Utilities- compressor drive system
- Utilities- instrument air system
- Utilities- start air system
- Utilities- diesel fuel system
- P&ID- glycol system - glycol flash tank
- P&ID- glycol system - glycol filters
- P&ID- glycol system - injection pumps
- P&ID- glycol system - lean / rich glycol exchanger
- P&ID- glycol system - glycol accumulator
- P&ID- glycol system - glycol regenerator
- P&ID- glycol system - glycol vapour condenser
- P&ID- glycol system - condensed water knock out
- P&ID- glycol system - condensed water pumps
- P&ID- glycol system- glycol sump tank
- P&ID- condensate surge vessel tie-ins
- P&ID- condensed water and produced water tie-ins
- P&ID- skimmer vessel & start up fuel gas tie-ins

- Utilities- fuel gas system
- Utilities- heat medium system
- Utilities- glycol system
- Facility block diagram
- Process data sheet
- All process project philosophies
- Study and investigation reports
- Input into Basis for design
- Process safeguarding memorandum narrative
- Process description
- Process system report
- Functional specifications and process data sheets for equipment
- Fluid flow and flow assurance design and operability report
- Relief, flare and blowdown study report

5.3. Piping / Mechanical Engineering

5.3.1. Definitions, Role and Responsibilities

Mechanical engineering discipline is responsible for the translation of process information into detailed equipment specification starting from generation of equipment list, equipment size confirmation, production of data sheets and other specifications. The mechanical discipline also deals with piping and other material specification taking cognisance of the service environment and other factors. Other key areas include equipment layout, stress analysis, painting specification and temperature and pressure rating specifications.

Piping design basis, piping gap analysis report, piping standards, piping mechanical calculations, data sheet for piping components, valve data sheet, piping material specifications and classes, general arrangement drawings, piping isometric drawings, material take-off, plot plans and equipment layout, pipe support list, pipe tie in drawings, installation workscope

Pipelines/Mechanical

Specifications for pigging facilities, river crossing drawings, mechanical data sheet for launchers/receivers, induction bend specifications, pipeline design basis, concrete coating calculations, crossing drawings, profile drawings, general layout route map, pipeline calculation notes, pipeline installation analysis and procedure, stability reports, gap analysis report, schedule of crossings, material take off.

Deliverables List for Mechanical Engineering Discipline

- Piping system design philosophy
- Pipeline design Philosophy
- Pipeline alignment sheets
- Pipe wall thickness calculations
- Buoyancy Calculations
- Plant layout philosophy
- Overall plant layout
- General piping & equipment layout
- Piping key plan
- Underground piping plan
- Facility plot plan
- 30% model review
- 60% model review
- 100% model review
- Piping General Arrangement Drawing - inlet manifold area
- Piping General Arrangement Drawing - inlet separators
- Piping General Arrangement Drawing - LTS area
- Piping General Arrangement Drawing - condensate feed drum
- Piping General Arrangement Drawing - condensate stabilizer tower & reboiler
- Piping General Arrangement Drawing - off-General Arrangement Drawings compression system
- Piping General Arrangement Drawing - sales General Arrangement Drawings metering system
- Piping General Arrangement Drawing - condensate metering & disposal system
- Piping General Arrangement Drawing - glycol regeneration area
- Piping General Arrangement Drawing - plant utility area
- Piping General Arrangement Drawing - flare area
- Piping General Arrangement Drawing - closed drain system
- Piping General Arrangement Drawing - fire water protection system
- Piping isometrics
- Piping materials specification
- Piping material MTO
- Pipes support MTO
- Piping MTO (tie-ins)
- Pipe support location drawing
- Pipe support details
- Piping support development procedure
- Tie-in schedule
- Piping stress analysis report
- Piping stress analysis procedure
- Construction work scope
- Manual valve list
- Preliminary layout methodology study report
- Isometrics (fabrication ISO's systems ISO's, stress ISO's)

- Datasheets – General Equipment
- Specifications for mechanical equipment
- Requisitions (pipe, fittings, valves, supports, special items)
- Construction specifications (fabrication, installation, flushing, cleaning, hydro-testing of pipe works)
- Final bulk MTO
- Pipe Stress Analysis Report
- Data sheet- stabilizer feed drum
- Data sheet- stabilizer feed bottom exchanger
- Data sheet- condensate stabilizer
- Data sheet- stabilizer bottom pumps
- Data sheet- general arrangement drawings/general arrangement drawings exchanger
- Data sheet- general arrangement drawings/liquid heat exchanger
- Data sheet- glycol regeneration package
- Data sheet- fuel general arrangement drawings scrubber
- Data sheet- start general arrangement drawings scrubber
- Data sheet -instrument/start air receiver
- Data sheet -plant air compressor
- Data sheet- fuel general arrangement drawings heater
- Data sheet- heat medium package
- Data sheet- heat medium system pumps
- Data sheet- fire water system pumps
- Data sheet- fire water tank
- Data sheet- off general arrangement drawings compressor
- Data sheet- heat medium surge tank
- Data sheet- HP flare knockout drum
- Data sheet- LP flare knockout drum
- Data sheet- test separator
- Driver selection study reports
- Input data to layout
- Noise/acoustic protection philosophies
- Noise control reports
- Mechanical handling reports
- Calculations
- Specifications and requisitions (pressure vessels, heat exchangers, tanks, packaged equipment)
- Final bulk MTO
- Rotating Equipment Type Selection report
- Heat Transfer Equipment Report
- Master Equipment List
- Mechanical Handling Study - Update
- Weld joint repair procedure
- Hydro-testing procedure
- Surface finishing and painting procedure

- Receiving inspection procedure
- Equipment calibration control procedure
- Control of nonconforming product procedure
- Corrective and preventive action procedure
- Material control procedure
- Welding material control procedure
- Welding procedure specification
- Welding tracking procedure
- Quality control procedure
- Vendor/supplier assessment procedure
- Technical query procedure
- Purchasing procedure
- Site clearing and excavation procedure
- Commissioning procedure
- Steel structural fabrication and erection procedure
- Piping fabrication and erection procedure
- Welding procedure
- Preservation and storage procedure
- Quality audit reports
- Inspection and test reports
- Site inspection reports
- Deviation and concession register
- ITP for mechanical works

5.4. Electrical Engineering

5.4.1. Definitions, Role and Responsibilities

Electrical engineering discipline deals with all details relating to equipment that run on electric power. It is responsible for power generation, distribution and use on site in a safe and proper manner so as to ensure that adequate electric power is calculated and provided for all equipment running on electric power.

Deliverables List for Electrical Engineering Discipline

- Electrical design philosophy
- General symbol and legend drawings
- Electrical load list (schedule)
- Cable sizing calculations
- Lighting calculations
- Fault level calculations
- Electrical overall cable route layout
- Electrical cable trays/ rack layouts
- Electrical equipment and switchboard rooms layout
- Electrical earthing general arrangement & layout
- Lighting & small power layouts
- Electrical key single line diagrams

- Electrical lighting layout diagram
- Hazardous area classification drawings
- Electrical cable tray/cable transit installation details
- LV switchboard data sheet
- Ups datasheet
- Data sheet - LV switchboard
- Data sheet - UPS
- Data sheet - electrical power generator
- Ups specification
- LV switchboard specification
- Generator specification
- Electrical material -take off (MTO)
- Electrical Earthing general arrangement
- Electrical installation work scope
- SAFOP report
- System Description - Electrical Design Philosophy
- Operation Philosophy
- Control, Monitoring & Protection Philosophy
- Emergency & Vital Supplies
- Power Generation and Distribution Study
- Short circuit
- Vital Supply Recovery Study
- Emergency & vital systems Single Line diagrams
- Protection & metering key diagrams
- LV switchgear schedule
- Summary of Electrical Engineering (electrical equipment summary sheets)
- Utility data (electrical)
- Specification for Power Management System
- Specification for Protection System
- Electrical Power and Protection System Study
- Electrical Utility Interface
- Electrical Equipment Data Sheets and Specifications

5.5. Instrumentation and Control Engineering

5.5.1. Definitions, Role and Responsibilities

Instrumentation and Controls engineering ensures that process equipment operate and are regulated to perform the various tasks they are designed to perform without any hitches. It ensures that equipment operation is controlled and regulated to guarantee the intended product from the plant.

Deliverables List for Instrumentation and Control Engineering Discipline

- Control, Monitoring & Protection Philosophy
- Emergency & Vital Supplies
- Power Generation and Distribution Study

- Short circuit
- Vital Supply Recovery Study
- Emergency & vital systems Single Line diagrams
- Protection & metering key diagrams
- LV switchgear schedule
- Summary of Electrical Engineering (electrical equipment summary sheets)
- Utility data (electrical)
- Specification for Power Management System
- Specification for Protection System
- Electrical Power and Protection System Study
- Electrical Utility Interface
- Electrical Equipment Data Sheets and Specifications
- Instrumentation and control design philosophy
- General specification for instrumentation
- Safety instrumented system specification
- Process controls narrative
- Process automation system specification
- Fire and gas system specification
- Instrument cable specifications
- Process automation system architectural drawing layout
- Cause and effect diagram matrix
- Emergency and safety shutdown system block diagram
- Fire and gas design study
- Fire and gas mapping study
- Fire and gas monitoring system(FGMS) layout diagram
- Fire and gas block diagram
- Metering system topology
- Metering system calculations
- Metering philosophy
- HMI & graphics philosophy
- Fieldbus planning tool philosophy
- Safety integrity level (SIL) classification report
- Instrument location diagram
- Instrument cable routing and junction box layout drawing
- Instrument and control systems block diagram
- Foundation fieldbus layout drawings
- Instrument air consumption calculations
- Cabinet power consumption and heat dissipation
- Instrument power supply distribution diagram
- Instrument air distribution diagram
- Control room layout and control cabinet location drawings
- Instrument index
- I / o, range trip, alarm and trip list
- Control valve sizing
- Instrument data sheets - pressure instruments

- Instrument data sheets - DP instruments
- Instrument data sheets - level instruments
- Instrument data sheets - flow instruments
- Instrument data sheets - temperature instruments
- Instrument data sheets - relief valves
- Instrument data sheets - control valves
- Instrument data sheets - on / off valves
- Instrument data sheets - orifices
- Instrument loop and termination diagrams
- Instrument cable schedule
- Instrument hook-up's drawings
- Instruments MTO
- Sis classification implementation report.
- Process automation system main distribution frame layout drawings.
- Process automation system cabinet layout drawings.
- Process automation system cabinet wiring diagrams
- Process automation system main distribution frame wiring diagrams
- Safety instrumented system main distribution frame layout drawings.
- Safety instrumented system cabinet layout drawings.
- Safety instrumented system main distribution frame wiring diagram
- Safety instrumented system cabinet wiring diagram
- Vendor package equipment unit control panel cabinet layout drawings
- Vendor package equipment unit control panel cabinet wiring diagram
- Interposing relay cabinet layout drawing
- Interposing relay cabinet wiring diagram
- Alarm rationalisation review report
- Foundation fieldbus calculation report
- ESD (HIS) panel layout drawing
- ESD (HIS)panel wiring diagram
- Specification for hot oil system
- Specification for instrument air system
- Specification for metering system
- Specification for instrument field auxiliary room
- Specification for all vendor equipment unit control panels
- SIS functional logic diagram
- Philosophy for the instrumentation of equipment packages
- Functional design specification for process automation system
- Functional design specification for safety instrumented system
- Functional design specification for fire and gas system
- Functional design specification for gas export metering system
- Functional design specification of vendor equipment unit control panel
- Control systems and safety instrumented systems interface diagrams
- SIS PFD calculation dossier
- FAT protocol for safety instrumented system
- FAT protocol for process automation system

- FAT protocol for vendor package unit control panel
- System integration test protocol for pas, sis and vendor equipment unit control panels.
- Site acceptance test protocol for safety instrumented system
- Site acceptance test protocol for safety instrumented system
- Site acceptance test protocol for vendor equipment unit control panel
- Cable layout and junction box details
- Loop drawings
- Fire and gas detectors hook-up drawings
- Functional logic diagram
- Fire and gas detection philosophy
- Detectors layout drawings
- Fire and gas mapping & study report
- FAT protocol
- Gas dispersion modelling report
- Fire and gas system main distribution frame layout drawings.
- Fire and gas system cabinet layout drawings.
- Fire and gas system main distribution frame wiring diagram
- Fire and gas system cabinet wiring diagram

5.6. Civil Engineering

5.6.1. Definitions, Role and Responsibilities

Civil engineering discipline ensures that equipment and structures are strong and solid enough to hold on during loading and idle period without failure. It ensures adequate foundation design and structural integrity for structures on site that carry the load of process and utility equipment. It covers other areas such as access road, drainage, paving and other essential civil engineering areas.

Deliverables List for Civil Engineering Discipline

- Civil/structural design philosophy (general)
- Geotechnical studies / reports
- Foundation design calculations report - inlet cooler skid
- Foundation design calculations report - inlet separator skid
- Foundation design calculations report - inlet manifold skid
- Foundation design calculations report - inlet separator
- Foundation design calculations report - test separator skid
- Foundation design calculations report - low temperature separator
- Foundation design calculations report - glycol regeneration Skid
- Foundation design calculations report - gas/gas exchanger skid
- Foundation design calculations report - booster compressor skid / building
- Foundation design calculations report - overhead compressor cooler skid
- Foundation design calculations report - overhead compressor skid
- Foundation design calculations report - stabilization feed drum
- Foundation design calculations report - stabilization reboiler skid
- Foundation design calculations report - fuel gas skid

- Foundation design calculations report - heat medium skid
- Foundation design calculations report - condensate stabilizer
- Foundation design calculations report - heat medium heater
- Foundation design calculations report - condensate product cooler
- Foundation design calculations report - LP flare knock out drum
- Foundation design calculations report - HP flare knock out drum
- Foundation design calculations report - firewater tank
- Foundation design calculations report - generator/utility building
- Structural design calculations report - flare tower
- Structural design calculations report - overhead compressor house
- Structural design calculations report - generator/utility building
- Foundation design calculations report – site offices
- Foundation design calculations report - gas/liquid exchanger
- Foundation design calculations report - control building
- Foundation design calculations report - pipe racks
- Foundation design calculations report - pipe supports
- Foundation drawings - inlet separator skid
- Foundation drawings - inlet manifold
- Foundation drawings - bulk inlet separator
- Foundation drawings - test separator skid
- Foundation drawings - low temperature separator
- Foundation drawings - gas / gas exchanger
- Foundation drawings - gas / liquid exchanger
- Foundation drawings - glycol regen.skid
- Foundation drawings - perimeter lighting poles
- Foundation drawings - condensate stabilizer
- Foundation drawings - stabilization feed drum
- Foundation drawings - stabilization reboiler skid
- Foundation drawings - overhead compressor cooler skid
- Foundation drawings - overhead compressor skid
- Foundation drawings - fuel gas skid
- Foundation drawings - heat medium skid
- Foundation drawings - condensate product cooler
- Foundation drawings - firewater tank
- Foundation drawings - bolster compressor / building
- Foundation drawings – site offices
- Foundation drawings - concrete sleepers
- Foundation drawings - pipe supports
- Structural drawings - overhead compressor house
- Structural drawings - flare tower
- Structural drawings - gate house
- Structural drawings - piperacks
- Structural drawings - generators/utility building
- Structural drawings - control building
- Detail drawings - pavings & drainages

- Detail drawings - fence/entrance gate (typical)
- Layout plan - pavings and drainages
- Bill of quantities /materials take off
- Construction work scope
- Site layout
- Fence, roads and drainage layout
- Foundation layout index
- Foundation layout areas
- Pipe rack layout and marking plan
- Specification for concrete foundation
- Specification for modules
- Specification for foundation
- Specification for steel
- Specification for water drainage and treatment
- Specification for drainage
- Specification for water supply
- Standard notes - civil engineering
- General notes for structural steel
- Structural bulk MTO report
- Piperack Isometrics
- Foundation, structural steel and concrete design
- Drainage, primary treatment and waste handling specifications
- Foundation Design or Anchor Systems Design
- Weight Management Estimate

5.7. Engineering Management

5.7.1. Introduction

Engineering management is a specialized form of management that is concerned with the application of engineering principles to business practice. It brings together the technological problem solving savvy of engineering and the organizational, administrative, and planning abilities of management in order to oversee complex enterprises from conception to completion. Engineering Management covers the following key areas:

- Work Methods and Procedures
- Design Review
- Safety in Design – HAZIP, HAZOP
- Value Engineering
- Constructability
- Engineering Design Verification
- Design Calculations
- Design Review
- Professional Authentication
- Equipment List And Numbering
- Instrumentation Index And Tagging

Project Execution Readiness
Engineering Requirements

- Line Numbering And Line List Development
- Data Sheet Development
- Preparation Of Master Calculations
- Regulatory Applications
- Drawing Status Index and Drawing Numbering
- Piping And Design Development
- Drawing Change Notice
- Process Flow Diagram (PFD) Development
- Piping and Instrumentation Diagram (P&ID) Development

6. Quality Assurance in Engineering Design

6.1. Introduction:

The Quality Management System for the project shall cover all project activities associated with the Engineering Design, Procurement, Installation, Construction, Commissioning of the Natural Gas Processing Plant and Gas Gathering and Transportation Pipelines. The quality assurance in engineering design will form part of the overall Project Quality Management System and will ensure that the engineering design activities are carried out in an error free, fit for purpose manner to achieve project objectives in accordance with the agreed standards and specifications.

6.2. General Requirements of the Project Quality Management System

- The project QMS shall be documented and maintained in order to continually improve its effectiveness according to the requirements of NIS - ISO 9001: 2000 international Standard. Engineering Design activities shall conform to the quality management system.
- The processes for the quality management system of engineering design activities shall be clearly identified in an engineering quality management manual and their applications in various Engineering Discipline Manuals, Procedures, and Work Method Statements.
- To establish the appropriate sequence and interaction of the processes, engineering disciplines shall identify own processes which though project specific, are clear cut representation of their operations and will be known as "Identified Processes" to be contained in the Engineering Quality Manual. These will then be detailed further in project files and other documentations.
- Management has the responsibility for providing the resources, the disbursement of which shall be based on individual project item budget requirements, to ensure that project quality objectives are effectively communicated and achieved.
- The effectiveness of engineering design quality processes is monitored, measured and analysed using our established quality assurance & quality control procedures and methods.
- Implementation of necessary actions geared towards achieving planned goals results from complying with the provisions of the project's design quality control methods. The data gathered in the form of results/experience, from project design engineering activities including feedback from internal (e.g. other disciplines) and external (e.g. vendors) clients shall form the basis for continual improvement of engineering design processes.

6.3. Design and Development

6.3.1. Design and Development Planning

Engineering design shall adequately be planned ahead and controlled during the actual design execution activity in development of processes, procedures, work methods, calculation sheets, drawings, specifications, reviews and other engineering activities. When necessary specialist skills not resident within the project team shall be outsourced to ensure that final output is of good quality and conforms to the project quality objectives.

Design and development stages are determined in line with the Basis for Design, Project Specifications, and other foundation documents. Usually for the size and complexity of the project, the design and development process will involve multiple stages.

Design development planning will determine the engineering reviews (e.g. Design Reviews, 30% Model reviews, and others), verification and validation that are appropriate to each design and development stage in line with the project quality requirements.

A dedicated engineering team led by a qualified and appropriately experienced Facilities Engineering Manager, who has hands-on experience in gas plant and/or Pipelines design shall handle all design and development activities, and it shall be the responsibility of the Facilities Engineering Manager to manage change control and ensure that the project engineering design requirements are met.

Effective communication shall be maintained between all relevant persons involved in the design and development planning stages of the project and this shall follow the project communication plan to ensure that problems associated with inadequate communication are reduced.

6.3.2. Design and Development Inputs

Design development requires some inputs relating to project requirements in order to fully understand the basis for which engineering solutions are being proposed. These required inputs shall be pre-determined and appropriately requested in line with approved process, and properly documented. Some inputs that will be required at the design stage of a gas plant include:

- a) Functional and Performance Requirement
- b) Applicable Statutory and Regulatory Requirements
- c) Necessary Design Codes and Standards
- d) Information obtained from previous similar designs in similar geographical location
- e) Other relevant requirements such as environmental and other pieces of information unique to the project location

To be assured of an optimal design output, these inputs are reviewed periodically to ensure consistency, alignment, adequacy, completeness and clarity by the engineering design team.

6.3.3. Design and Development Outputs

These outputs form the deliverables list detailed in section 5 of this document as well as the basis on which further work by other disciplines or expertise are performed .

The output requirements to be met by the deliverables from the engineering disciplines will:

- Form the input requirements for design and development
- Provide appropriate information for downstream activities such as detailed design, and procurement
- Contain or reference design acceptance criteria which in turn flows from the project objectives
- Specify the requirements of the project necessary for a safe and proper operation
- Confirm that design deliverables shall be approved as per project quality specifications before use.

6.3.4. Design and Development Review

All designs shall be subject to squad checks at intervals, with originator, checker and approval authority personnel clearly identified. This process will be followed for IFR, IFD and IFC stages of all documents including drawings and reports.

7. Safety Systems in Engineering Design

7.1. Introduction

Safety of people, protection of the environment and maintenance of asset integrity shall be an integral part of engineering design on the project from the onset. Where possible, potential for accidents, incidents and failures shall be identified, reviewed and eliminated at the design stage. This is the surest way of eliminating unwanted incidents.

The Project HSE Case shall maintain, within the project's overall work activities, plans for achieving HSE objectives and performance criteria. These plans include:

- A clear description of the plan's objectives
- Responsibility for setting and achieving the objectives
- Performance acceptance criteria for each distinct work package
- The means by which they are to be achieved
- Resource requirements for achievement of objectives
- Time scales for implementation of work activities that lead to the attainment of objectives

These plans are to be supported by:

- Programmes for motivating and encouraging personnel toward a suitable HSE culture
- Mechanisms to provide feedback to personnel on HSE performance
- Processes to recognise good personal and team HSE performance (e.g. safety award schemes)
- Mechanism for evaluation of HSE performance and a planned follow-up
- These plans shall form an integral part of the project management process for the natural gas processing plant and associated pipelines

7.2. HSE Planning in Engineering Design

HSE planning in engineering design involves a formal process used to formulate and disseminate the project HSE plans and targets among design engineering personnel such that mechanisms for achieving those targets form an integral part of the design. The HSE plan covers the lifespan of the project and contains in detail actions to be executed at any given time frame or stage of work. The Plan is owned and approved by the project HSE manager but execution is by engineering design teams.

Supporting the project HSE Plan are lower level operational HSE Plans among the engineering disciplines. These are developed by the various disciplines of work groups following issuance of the Project HSE Case. They are supportive and consistent with the Project HSE case. Discipline leads are free to integrate in to their HSE Plans any HSE

action they consider necessary at their level to ensure the Project HSE Policy and Objectives are met at the stage of work they manage.

Performance against the Plan is reviewed at every Project HSE Steering Committee. Discipline leads review performance against their own Plans at their own HSE meetings. The HSE Manager monitors all performance against plans.

7.3. Procedures and Work Instructions

Procedures describe the action that must be undertaken in order to meet policy and objectives. Engineering design operations maintain work procedures in addition to HSE Procedures. These may be grouped Work Instructions, Specifications, and Method Statements.

Compliance to these Procedures should ensure the Project HSE Case is executed in a uniform manner throughout the entire life and breath of the project. Procedures are supported by guidelines. These provide additional information to enable Managers and Staff to implement procedures. Guidelines are not mandatory. The HSE Manager controls all Project HSE. Discipline Leads can develop local guidelines if considered necessary.

Documented HSE procedures which have direct bearing on HSE management shall be brought to the attention of new personnel, during their induction.

At discipline level, work instructions shall be used to communicate hazards and controls and strategies to manage them. Such work instructions are derived from HAZID, HRA, EIA, Job Hazard Analysis, Permit-To-Work conditions and recommendations from reviews of previous incidents. They are communicated to the workforce through the Worksite Hazard Information System (WHIS). In addition toolbox talks, site HSE meetings, and briefing cards can be used to communicate procedures to project staff.

7.4. Emergency Response Plan

Separate procedures shall be put in place for Emergency Response at both the project and discipline levels. Details of Project Emergency Response provisions shall be made available to all staff on the project intranet website and other platforms where the following shall also be accessible:

- Project Emergency Response Contact telephone numbers
- Project Emergency Response Duty rosters
- Project Emergency Response policy
- Project Emergency Response Action checklist
- Project Contingency Plans and Procedures
- Project Major Incident Contingency Plans and Procedures

7.5. Integrity of Assets

Natural Gas processing facilities and associated pipeline assets shall be designed and managed for asset integrity to ensure acceptable HSE performance during operations phase. Both the Technical and Operating Integrity requirements shall be established early in the design phase, subsequently verified and maintained afterwards, by so doing the HSE performance of the asset should be within acceptable range provided correct design criteria have been followed. The responsibility for asset design integrity management is vested in the project team. Project team shall document the application of standards; review procedures, change management, and hazards and effects management tools as part of asset integrity management process and these shall take place at the planning and design stages. This strategy assures the owner of the integrity of the asset when completed and ready to be put into operations.

7.6. Design Change Control Process

The Change Control Process is used to manage modifications to approved designs or the incorporation of changes that came to light after approvals. This process is necessary to preserve asset integrity and to ensure that subsequent activities are based on the new change. Through this process, any proposed alterations to an approved design are evaluated and approved at an appropriate level and any potential consequence understood and prepared for prior to the change being effected. In this way, changes are properly engineered, reviewed, executed and documented. Fundamental aspects of the change control procedure are:

- I. The implications of change on all aspects of technical and operating integrity have been considered and
- II. Mutually consistent engineering documents record the change.

This process requires that the impact of any proposed change be thoroughly evaluated by a technically competent body or an engineering review team and approved by the project manager.

8. Engineering Design Organizational Structure

Figure 1 below shows a typical engineering design organization geared towards executing engineering design for the size and complexity of a natural gas plant, gas gathering and evacuation facilities.

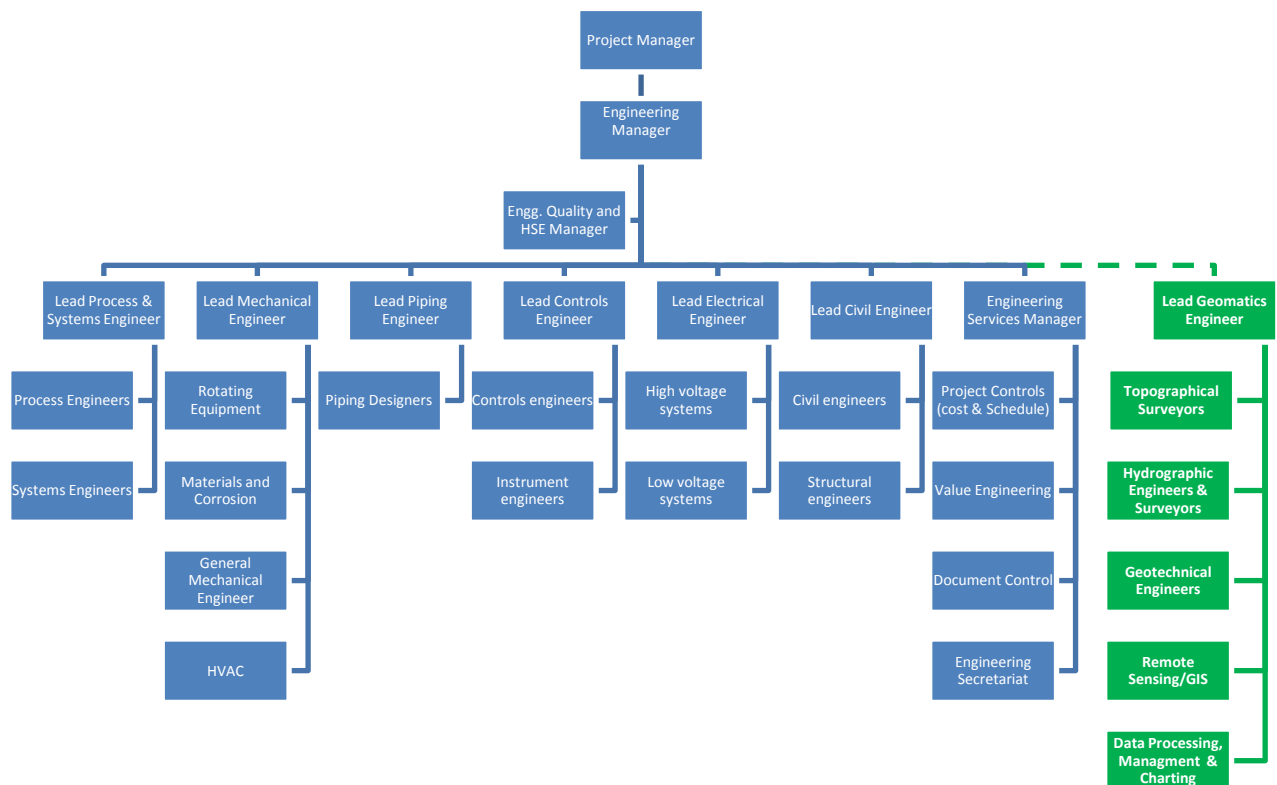
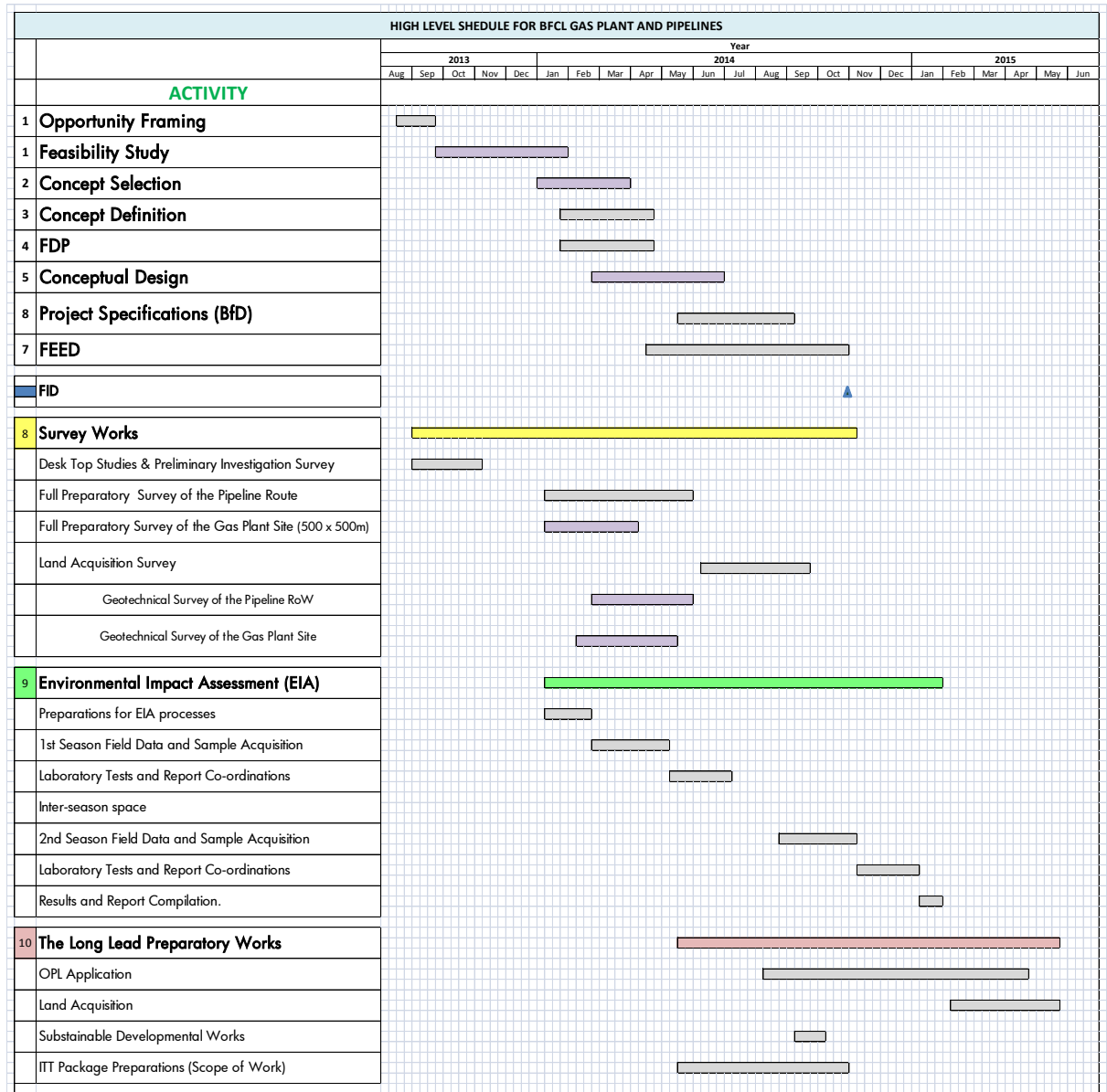


Figure 1 – Engineering Design Organizational Chart

9. Preliminary Pre-FID Activities Schedule

A high level schedule for the Pre-FID activities is shown, covering engineering & related activities, and long lead preparatory works that must be started on time to avoid project delay.



10. High Level Pre-FID Cost Estimate

The following tables provide a high level cost estimate at +/-30% accuracy level, covering Engineering & Related Activities

Gas Plant & Pipelines

S/N	Activity	Cost \$MM	Remarks
1	Feasibility Study	0.20	
2	Concept Selection	0.10	
3	Concept Definition	0.10	
4	FDP	0.25	
5	Conceptual Design – including Project Specification (BfD)	3.00	
6	FEED	7.00	
Survey Works			
7	Desktop Studies for PI, - for PTS Application	0.05	
8	FP Survey for OPL Application and P/L Route Land Acquisition	0.34	
9	Geomatics Support for P/L Route Land Acquisition	0.05	P/L Route Only. Gas Plant Site will be covered separately
10	Geotechnical Survey – P/L Route	0.20	“
EIA			
11	EIA	0.28	
12	Provision for Long Lead Preparatory Works	0.10	
	Total	11.67	